

FABIAN VAZQUEZ

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EDUCATION

The University of Texas Rio Grande Valley M.S. Computer Science	GPA = 3.67	Expected Graduation: August 2024
The University of Texas Rio Grande Valley B.S. Mechanical Engineering	GPA = 3.77	Graduation: May 2022

Technical Skills

- Programming Languages: Python (2 years), C++ (3 months)
- Frameworks: Flask (1 year)
- Web Technologies: HTML (1 year), CSS (1 year)
- Cloud Platforms: Google Cloud (1 year)
- Tools: YOLO, MPI, CUDA, GitHub, Google Cloud Run, Cloudflare, NGINX, Docker
- Languages: English (Fluent), Spanish (Fluent)

RELEVANT EXPERIENCE

College of Engineering and Computer Science UTRGV Edinburg, TX <i>Graduate Research Assistant</i>	June 2023 - Present
• Conducting advanced research on deep learning and computer vision techniques to enhance polyp detection from colonoscopy images • Utilize YOLO (You Only Look Once) models to develop and optimize object detection algorithms for medical imaging applications	
College of Engineering and Computer Science UTRGV Edinburg, TX <i>Teaching Assistant, Parallel Computing</i>	January 2024 - May 2024
• Designed parallel programming projects using MPI and CUDA • Provided student support through email and in-class guidance	
Colorado Engineering Inc Colorado Springs, CO <i>Engineering Intern</i>	June - August 2021
• Streamlined copper wire assembly process by designing and implementing 3D printed templates, reducing prototype assembly time by 50% • Soldered and assembled high-frequency RF semi-rigid copper wires, ensuring precision and reliability	

ACADEMIC EXPERIENCE

Polypdetector , Thesis Research Project	August 2023 - Present
• Training and optimizing YOLO models for polyp detection in colonoscopy images • Developed a Flask web interface for user interaction and image upload • Implementing a continuous integration and continuous deployment (CI/CD) pipeline using GitHub and Google Cloud, automating the deployment process on Google Cloud Run	
CUDA and MPI Project , Parallel Computing Class Project	March - May 2024
• Reduced runtime of complex mathematical operations using MPI and parallel computing by up to 123 times faster • Implemented CUDA to compare execution speeds and identify the fastest runtimes for various computations	
Catdetector , Summer Research Project	June - August 2023
• Developed and trained a YOLO model to accurately detect cats in images • Built a flask web application using Python, HTML, and CSS to allow users to test images for cats • Deployed the Flask app on Google Cloud Platform's Compute Engine	

LEADERSHIP AND ACTIVITIES

Kokua Education - IDEA Public Schools South Texas, TX <i>Substitute Teacher</i>	December 2022 - May 2023
• Filled in for absent teachers across all subjects and grade levels (K-12) • Maintained student engagement and motivation through energetic and enthusiastic teaching	
Extracurricular Activities	
• <i>Brazilian Jiu Jitsu</i>: Practicing jiu jitsu for 1.5 years • <i>Reading</i>: Passionate reader with a focus on sci-fi and classic fiction books • <i>Weight Lifting</i>: Have enjoyed regular weight lifting for 3 years	